MSHA Finger Pinch/Crush Narrative Comparison

K-means Cluster Compared with the Closest LDA Topic

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Narratives analyzed: 2,000
Shared vocabulary terms: 2,763
K-means clusters: 9
K-means silhouette score: 0.0533
Selected finger cluster: 6
Selected cluster size: 247
LDA topics: 9
Matched LDA topic: 3
Agreement records: 127
K-means-only records: 120
LDA-only records: 120
Document-set Jaccard: 0.346

Purpose:
Assess whether a first-pass text cluster contains the expected finger
pinch, crush, caught-between, and amputation narratives, and determine
whether the related LDA topic identifies relevant missing documents.
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Executive Summary

This analysis compares one K-means document cluster with the most closely related LDA topic using the same 2,000 severity-balanced MSHA narratives and synchronized vocabulary preprocessing.

K-means cluster 6 was automatically selected because its vocabulary and document content had the strongest finger, pinch, crush, caught-between, degloving, and amputation pattern.

The selected cluster contained 247 narratives. Its mean document-level silhouette value was 0.092, and 0 records had negative silhouette values.

LDA topic 3 was selected as the closest topic because its document proportions were concentrated within the selected K-means cluster and its highest-proportion document set had the greatest overlap with that cluster.

The methods agreed on 127 records. K-means included 120 records that were not in the strongest LDA set, while LDA identified 120 records from outside the K-means cluster.

K-means forces each narrative into one cluster. LDA instead gives each narrative a mixture of topic proportions. Therefore, the methods are not expected to produce identical document sets.

Synchronized Methodology

1. The same severity-balanced sampling logic from the prior clustering analysis was used to select 2,000 narratives.

2. The same narrative normalization and stop-word rules were applied.

3. One CountVectorizer created a shared corpus vocabulary containing individual words and two-word phrases.

4. For K-means, the count matrix was converted to TF-IDF, reduced with Truncated SVD, and grouped into nine clusters.

5. For LDA, the original count matrix was used directly. LDA represented every narrative as a mixture of nine topic proportions.

6. The finger pinch/crush K-means cluster was selected automatically from its top terms and the frequency of finger-related narratives.

7. The closest LDA topic was determined using topic concentration inside the selected cluster, separation from outside records, top-document overlap, enrichment, and Jaccard similarity.

8. The strongest LDA document set was defined as the same number of highest-proportion records as the size of the K-means cluster.

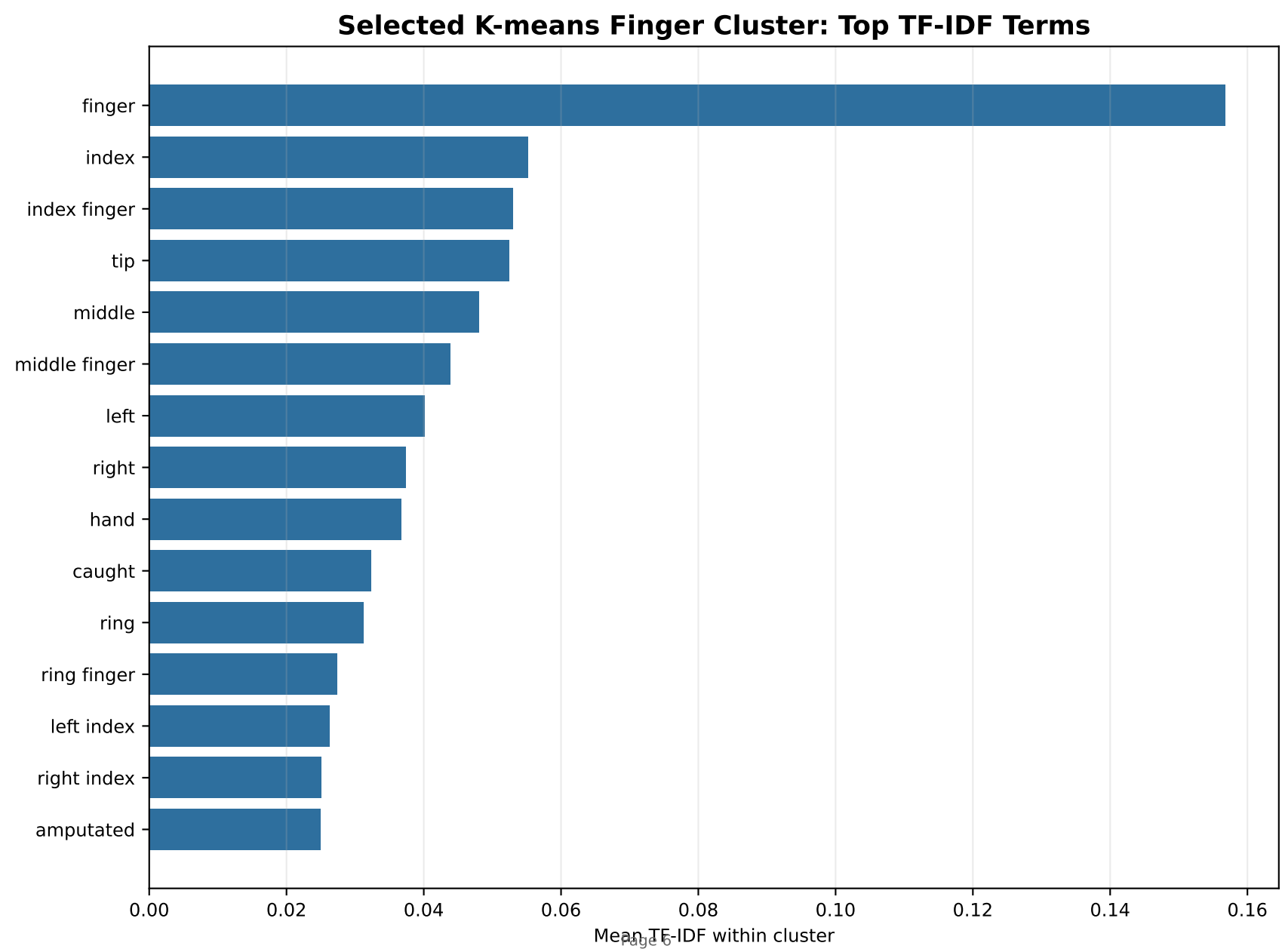
9. Existing MSHA fields were retained only for interpretation and were not used to create the cluster or topic.

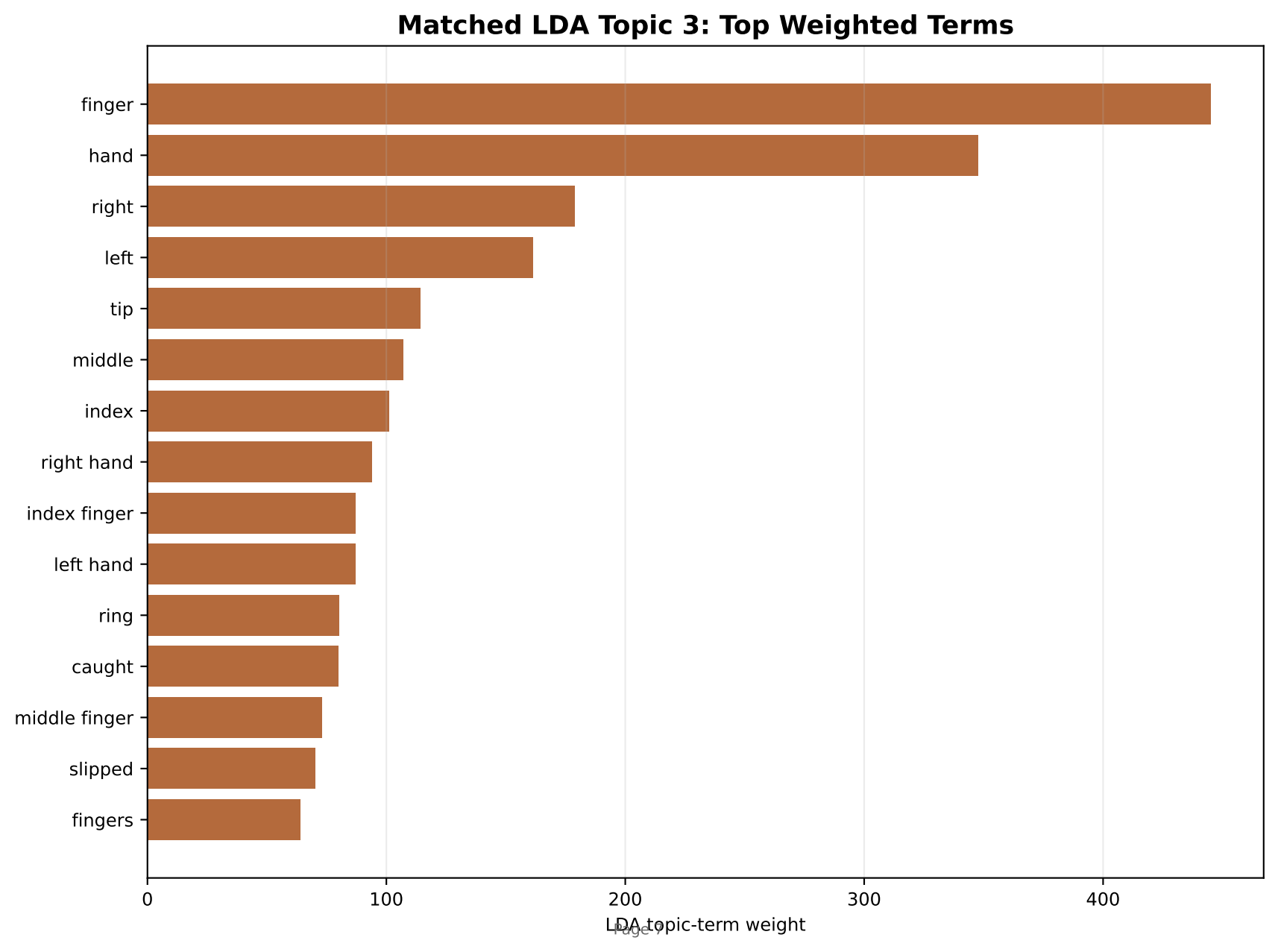
Severity-Stratified Sample

Severity Group	Available Records	Target	Initially Selected	Final Selected
Fatality	1188	400	400	400
Permanent disability	2455	400	400	400
Days away	178432	400	400	400
Restricted or transferred	42788	400	400	400
No lost time or other	43965	400	400	400

Finger Cluster Selection Scores

Cluster Id	Vocabulary Score	Finger-Event Narrative Rat	Combined Selection Score	Cluster Size
6	18.3671	1.0000	33.3671	247
8	4.7820	0.4274	11.1933	248
3	0.0000	0.0640	0.9600	125
1	0.0000	0.0526	0.7895	114
5	0.0000	0.0347	0.5202	346
7	0.0000	0.0338	0.5072	414
4	0.0000	0.0213	0.3191	188
0	0.0000	0.0114	0.1705	176
2	0.0000	0.0000	0.0000	142


Selected K-means Finger Cluster: Top TF-IDF Terms
finger
index
index finger
tip
middle
middle finger
left
right
hand
caught
ring
ring finger
left index
right index
amputated
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Mean TF-IDF within cluster


Matched LDA Topic 3: Top Weighted Terms
finger
hand
right
left
tip
middle
index
right hand
index finger
left hand
ring
caught
middle finger
slipped
fingers
0
100
200
300
400
LDA topic-term weight

Vocabulary Comparison

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SHARED TOP TERMS
--------------------------------------------------------------------------------
caught, finger, hand, index, index finger, left, middle, middle finger, right, ring, tip

K-MEANS-ONLY TOP TERMS
--------------------------------------------------------------------------------
amputated, left index, right index, ring finger

LDA-ONLY TOP TERMS
--------------------------------------------------------------------------------
fingers, left hand, right hand, slipped

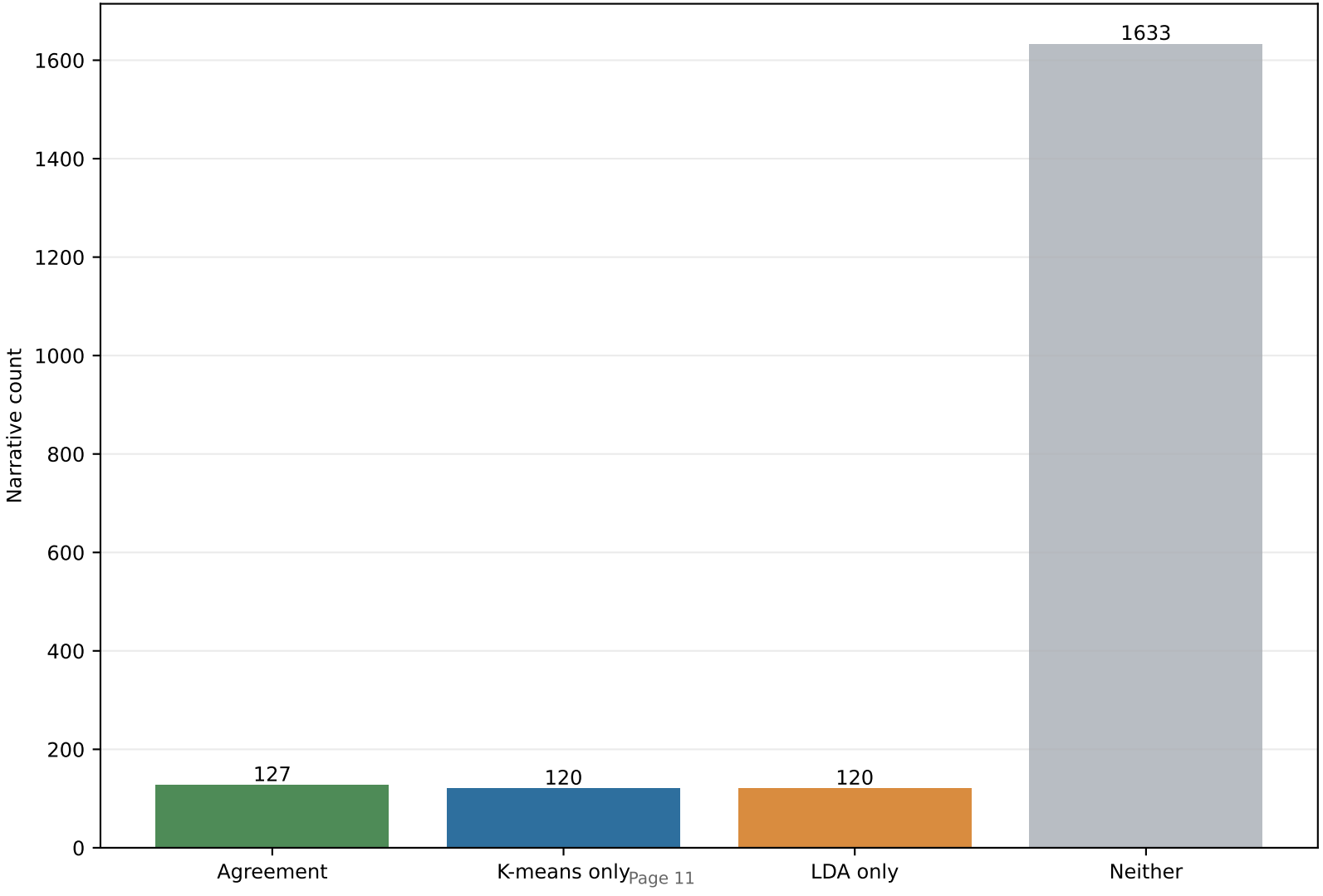
INTERPRETATION
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K-means terms describe vocabulary that distinguishes one document
cluster from the rest of the corpus. LDA terms describe words with
high weight in a probabilistic topic that may appear to different
degrees across many narratives.
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Top-Term Overlap Summary

Measure	Value
K-means top terms	15.0000
LDA top terms	15.0000
Shared top terms	11.0000
Top-term Jaccard	0.5789

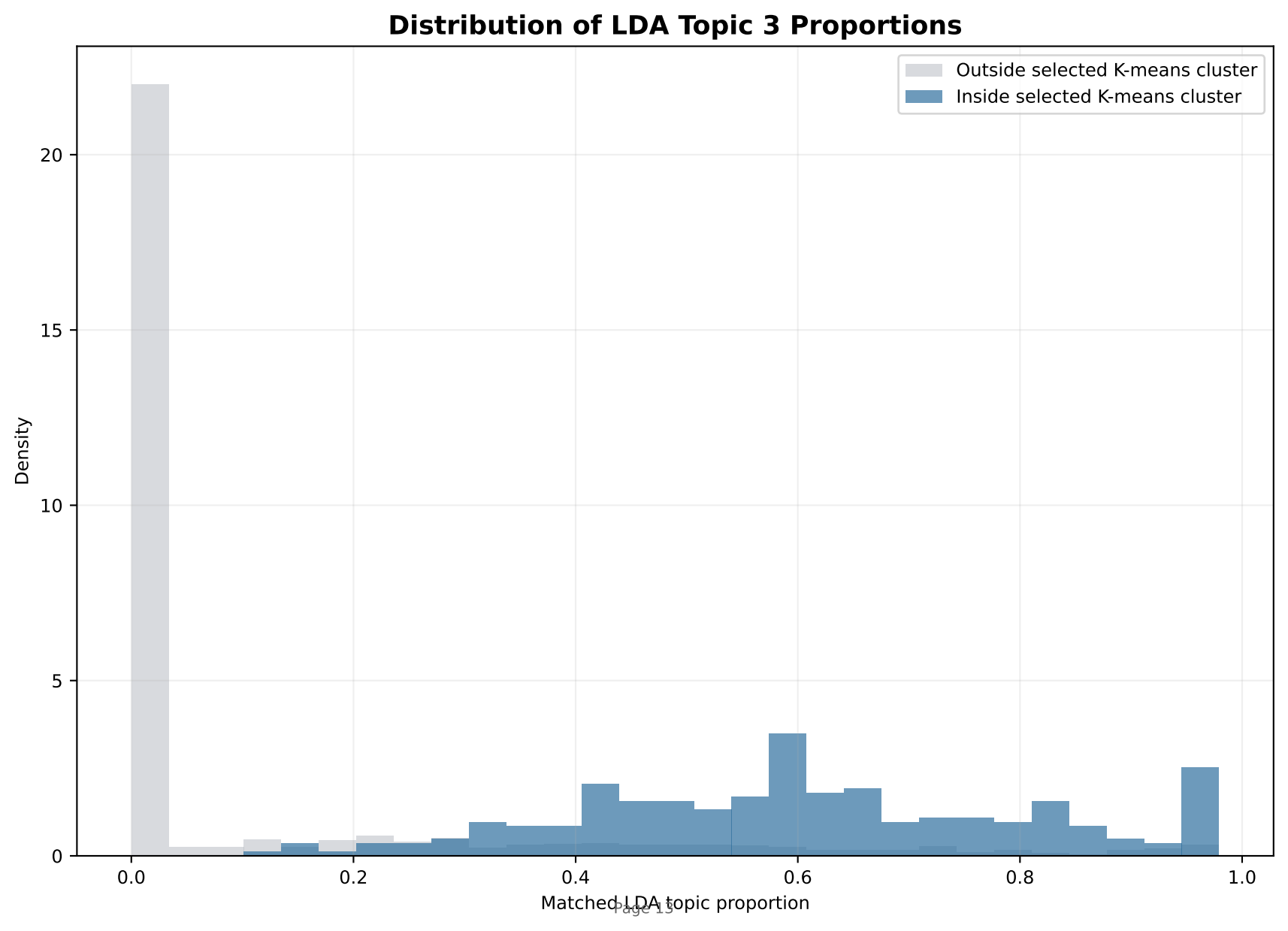
LDA Topic Matching Results

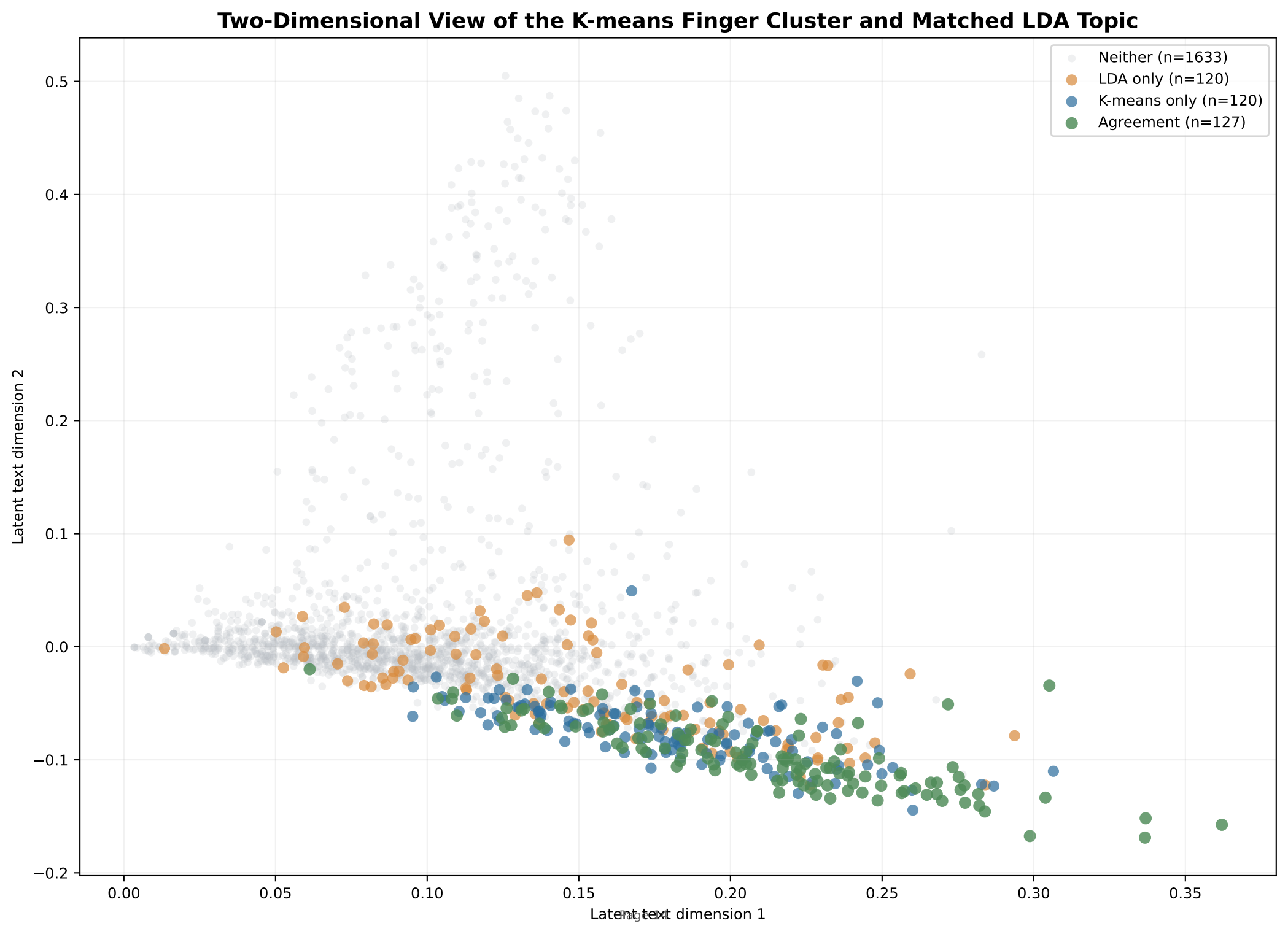
Topic Id	Mean Proportion Insi	Mean Proportion Out	Inside-Outside Separ	Enrichment Ratio	Top-N Overlap	Cluster Recall	Jaccard Similarity	Combined Match Score
3	0.6069	0.1165	0.4904	5.2095	127	0.5142	0.3460	0.5649
8	0.1373	0.1326	0.0047	1.0357	23	0.0931	0.0488	0.0935
2	0.0729	0.1434	-0.0706	0.5081	8	0.0324	0.0165	0.0286
1	0.0496	0.1040	-0.0545	0.4764	14	0.0567	0.0292	0.0240
7	0.0400	0.0817	-0.0417	0.4894	16	0.0648	0.0335	0.0240
6	0.0393	0.1175	-0.0782	0.3343	4	0.0162	0.0082	0.0064
5	0.0123	0.0791	-0.0667	0.1561	3	0.0121	0.0061	-0.0068
0	0.0284	0.1340	-0.1056	0.2122	1	0.0040	0.0020	-0.0089
4	0.0133	0.0912	-0.0779	0.1456	4	0.0162	0.0082	-0.0090


K-means Cluster and LDA Topic Document Comparison
Narrative count
0
200
400
600
800
1000
1200
1400
1600
127
120
120
1633
Agreement
K-means only
LDA only
Neither

Document Overlap Measures

Measure	Value
Agreement records	127.0000
K-means-only records	120.0000
LDA-only records	120.0000
LDA precision relative to K-means	0.5142
LDA recall of K-means cluster	0.5142
Document-set Jaccard	0.3460


Distribution of LDA Topic 3 Proportions
Outside selected K-means cluster
Inside selected K-means cluster
Density
0
5
10
15
20
0.0
0.2
0.4
0.6
0.8
1.0
Matched LDA topic proportion


Two-Dimensional View of the K-means Finger Cluster and Matched LDA Topic
Neither (n=1633)
LDA only (n=120)
K-means only (n=120)
Agreement (n=127)
Latent text dimension 1
Latent text dimension 2
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2

Post-Hoc Existing-Field Comparison

The structured MSHA fields shown in the following table were not used to build either the K-means cluster or the LDA topic.

The comparison is used only to understand whether Agreement, K-means-only, and LDA-only records share similar existing classifications, accident types, activities, injury sources, injury natures, body parts, and severity levels.

Most Common Existing Values by Comparison Group

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
Agreement	CLASSIFICATION	1	HANDLING OF MATERIALS	59	46.5%
Agreement	CLASSIFICATION	2	MACHINERY	36	28.3%
Agreement	CLASSIFICATION	3	POWERED HAULAGE	14	11.0%
Agreement	CLASSIFICATION	4	HANDTOOLS (NONPOWERED)	14	11.0%
Agreement	CLASSIFICATION	5	SLIP OR FALL OF PERSON	4	3.1%
Agreement	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	51	40.2%
Agreement	ACCIDENT_TYPE	2	Caught in, under or between NEC	34	26.8%
Agreement	ACCIDENT_TYPE	3	Caught in, under or between running or meshing objects	12	9.4%
Agreement	ACCIDENT_TYPE	4	Struck by... NEC	12	9.4%
Agreement	ACCIDENT_TYPE	5	Struck by falling object	8	6.3%
Agreement	ACTIVITY	1	Machine maintenance	34	26.8%
Agreement	ACTIVITY	2	Handling supplies or material	32	25.2%
Agreement	ACTIVITY	3	Hand tools (not powered)	15	11.8%
Agreement	ACTIVITY	4	Roof bolter, NEC	6	4.7%
Agreement	ACTIVITY	5	Handling coal, Rock, Ore	6	4.7%
Agreement	INJURY_SOURCE	1	METAL,NEC(PIPE,WIRE,NAIL)	23	18.1%
Agreement	INJURY_SOURCE	2	METAL COVERS & GUARDS	19	15.0%
Agreement	INJURY_SOURCE	3	BELT CONVEYORS	8	6.3%
Agreement	INJURY_SOURCE	4	UNDERGRD MINING MACHINES	7	5.5%
Agreement	INJURY_SOURCE	5	BROKEN ROCK,COAL,ORE,WSTE	7	5.5%
Agreement	NATURE_INJURY	1	AMPUTATION OR ENUCLEATION	95	74.8%
Agreement	NATURE_INJURY	2	CUT,LACER,PUNCT-OPN WOUND	14	11.0%
Agreement	NATURE_INJURY	3	FRACTURE,CHIP	12	9.4%

Most Common Existing Values by Comparison Group - Continued

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
Agreement	NATURE_INJURY	4	CRUSHING	4	3.1%
Agreement	NATURE_INJURY	5	DISLOCATION	1	0.8%
Agreement	INJ_BODY_PART	1	FINGER(S)/THUMB	123	96.9%
Agreement	INJ_BODY_PART	2	HAND (NOT WRIST OR FINGERS)	4	3.1%
Agreement	DEGREE_INJURY	1	PERM TOT OR PERM PRTL DISABLTY	96	75.6%
Agreement	DEGREE_INJURY	2	DAYS RESTRICTED ACTIVITY ONLY	16	12.6%
Agreement	DEGREE_INJURY	3	NO DYS AWY FRM WRK,NO RSTR ACT	11	8.7%
Agreement	DEGREE_INJURY	4	DAYS AWAY FROM WORK ONLY	4	3.1%
K-means only	CLASSIFICATION	1	HANDLING OF MATERIALS	62	51.7%
K-means only	CLASSIFICATION	2	MACHINERY	23	19.2%
K-means only	CLASSIFICATION	3	HANDTOOLS (NONPOWERED)	19	15.8%
K-means only	CLASSIFICATION	4	POWERED HAULAGE	11	9.2%
K-means only	CLASSIFICATION	5	SLIP OR FALL OF PERSON	4	3.3%
K-means only	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	53	44.2%
K-means only	ACCIDENT_TYPE	2	Caught in, under or between NEC	26	21.7%
K-means only	ACCIDENT_TYPE	3	Struck by... NEC	16	13.3%
K-means only	ACCIDENT_TYPE	4	Struck by falling object	10	8.3%
K-means only	ACCIDENT_TYPE	5	Caught in, under or between running or meshing objects	7	5.8%
K-means only	ACTIVITY	1	Machine maintenance	44	36.7%
K-means only	ACTIVITY	2	Handling supplies or material	31	25.8%
K-means only	ACTIVITY	3	Hand tools (not powered)	18	15.0%
K-means only	ACTIVITY	4	Handling coal, Rock, Ore	4	3.3%
K-means only	ACTIVITY	5	Hand tools (powered)	3	2.5%

Most Common Existing Values by Comparison Group - Continued

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
K-means only	INJURY_SOURCE	1	METAL,NEC(PIPE,WIRE,NAIL)	31	25.8%
K-means only	INJURY_SOURCE	2	METAL COVERS & GUARDS	16	13.3%
K-means only	INJURY_SOURCE	3	HAND TOOLS,POWERED,NEC	7	5.8%
K-means only	INJURY_SOURCE	4	BROKEN ROCK,COAL,ORE,WSTE	6	5.0%
K-means only	INJURY_SOURCE	5	AXE,HAMMER,SLEDGE	6	5.0%
K-means only	NATURE_INJURY	1	AMPUTATION OR ENUCLEATION	72	60.0%
K-means only	NATURE_INJURY	2	CUT,LACER,PUNCT-OPN WOUND	25	20.8%
K-means only	NATURE_INJURY	3	FRACTURE,CHIP	14	11.7%
K-means only	NATURE_INJURY	4	MULTIPLE INJURIES	3	2.5%
K-means only	NATURE_INJURY	5	CRUSHING	3	2.5%
K-means only	INJ_BODY_PART	1	FINGER(S)/THUMB	120	100.0%
K-means only	DEGREE_INJURY	1	PERM TOT OR PERM PRTL DISABLTY	72	60.0%
K-means only	DEGREE_INJURY	2	DAYS RESTRICTED ACTIVITY ONLY	26	21.7%
K-means only	DEGREE_INJURY	3	NO DYS AWY FRM WRK,NO RSTR ACT	19	15.8%
K-means only	DEGREE_INJURY	4	DAYS AWAY FROM WORK ONLY	1	0.8%
K-means only	DEGREE_INJURY	5	DYS AWY FRM WRK & RESTRCTD ACT	1	0.8%
LDA only	CLASSIFICATION	1	MACHINERY	37	30.8%
LDA only	CLASSIFICATION	2	HANDLING OF MATERIALS	34	28.3%
LDA only	CLASSIFICATION	3	POWERED HAULAGE	21	17.5%
LDA only	CLASSIFICATION	4	HANDTOOLS (NONPOWERED)	16	13.3%
LDA only	CLASSIFICATION	5	FALL OF ROOF OR BACK	6	5.0%
LDA only	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	43	35.8%
LDA only	ACCIDENT_TYPE	2	Struck by... NEC	25	20.8%

Most Common Existing Values by Comparison Group - Continued

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
LDA only	ACCIDENT_TYPE	3	Struck by falling object	19	15.8%
LDA only	ACCIDENT_TYPE	4	Caught in, under or between NEC	11	9.2%
LDA only	ACCIDENT_TYPE	5	Caught in, under or between running or meshing objects	11	9.2%
LDA only	ACTIVITY	1	Machine maintenance	34	28.3%
LDA only	ACTIVITY	2	Handling supplies or material	15	12.5%
LDA only	ACTIVITY	3	Hand tools (not powered)	11	9.2%
LDA only	ACTIVITY	4	Roof bolter, NEC	10	8.3%
LDA only	ACTIVITY	5	Continuous miner	6	5.0%
LDA only	INJURY_SOURCE	1	UNDERGRD MINING MACHINES	19	15.8%
LDA only	INJURY_SOURCE	2	METAL,NEC(PIPE,WIRE,NAIL)	15	12.5%
LDA only	INJURY_SOURCE	3	BELT CONVEYORS	10	8.3%
LDA only	INJURY_SOURCE	4	MINE JEEP,KERSEY,JITNEY	8	6.7%
LDA only	INJURY_SOURCE	5	METAL COVERS & GUARDS	8	6.7%
LDA only	NATURE_INJURY	1	AMPUTATION OR ENUCLEATION	49	40.8%
LDA only	NATURE_INJURY	2	CUT,LACER,PUNCT-OPN WOUND	18	15.0%
LDA only	NATURE_INJURY	3	CRUSHING	18	15.0%
LDA only	NATURE_INJURY	4	FRACTURE,CHIP	18	15.0%
LDA only	NATURE_INJURY	5	MULTIPLE INJURIES	7	5.8%
LDA only	INJ_BODY_PART	1	FINGER(S)/THUMB	67	55.8%
LDA only	INJ_BODY_PART	2	HAND (NOT WRIST OR FINGERS)	12	10.0%
LDA only	INJ_BODY_PART	3	MULTIPLE PARTS (MORE THAN ONE MAJOR)	8	6.7%
LDA only	INJ_BODY_PART	4	TRUNK, MULTIPLE PARTS	6	5.0%
LDA only	INJ_BODY_PART	5	FOOT(NOT ANKLE/TOE)/TARSUS/METATARSUS	4	3.3%

Most Common Existing Values by Comparison Group - Continued

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
LDA only	DEGREE_INJURY	1	PERM TOT OR PERM PRTL DISABLTY	50	41.7%
LDA only	DEGREE_INJURY	2	NO DYS AWY FRM WRK,NO RSTR ACT	21	17.5%
LDA only	DEGREE_INJURY	3	DAYS RESTRICTED ACTIVITY ONLY	21	17.5%
LDA only	DEGREE_INJURY	4	FATALITY	18	15.0%
LDA only	DEGREE_INJURY	5	DAYS AWAY FROM WORK ONLY	9	7.5%

Agreement: Documents Identified by Both Methods - 1 of 2

These narratives are members of the selected K-means cluster and also rank among the strongest documents for the matched LDA topic. They provide the clearest shared interpretation of the cluster and topic.

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Example 1 | Record 220002310036 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.160
Matched LDA topic proportion: 0.697 | Dominant LDA topic: 3
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between NEC

ON SOUTH END HALF DECK OF PRESS FLOOR, TAKING DOWN AN AIR BLOW DOWN POT, WHEN IT ROLLED AND JAMMED HIS RIGHT INDEX FINGER AGAINST THE MID BAR OF THE HAND RAIL. DISTAL TIP OF FINGER AMPUTATED.

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Example 2 | Record 220112910049 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.182
Matched LDA topic proportion: 0.951 | Dominant LDA topic: 3
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

He was helping lay track and caught his right index finger and middle finger between the bottom of a track tie and the bottom and cut the tips off each finger.

==
Example 3 | Record 220171160040 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.164
Matched LDA topic proportion: 0.702 | Dominant LDA topic: 3
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

Employee was putting a bearing on a mixer with hand. The bearing moved quicker then expected and ee smashed finger between the bearing and the mixer. The tip of left index finger was amputated by dr.

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Example 4 | Record 220213410008 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.143
Matched LDA topic proportion: 0.886 | Dominant LDA topic: 3
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

Employee was cleaning inside of a Reed Gunite Pot and got left hand glove caught between feed wheel and rock shear. Severed tip of middle finger and index finger.

==
Example 5 | Record 220120230023 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.178
Matched LDA topic proportion: 0.597 | Dominant LDA topic: 3
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

Moving 24-10 915 Eimco out of surface diesel shop, pinched left index finger between steering ram and

Agreement: Documents Identified by Both Methods - 2 of 2

hydrlic hose. Laceration and smashed finger this resulted in amputation of finger tip below the first joint.

==
Example 6 | Record 220130880012 | Days away
K-means cluster: 6 | K-means silhouette: 0.174
Matched LDA topic proportion: 0.647 | Dominant LDA topic: 3
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between NEC

EMPLOYEE WAS LOADING A TOP ROLLER ONTO A SKID STEER WHILE TEARING DOWN OLD BELT LINE AN SMASHED HIS LEFT MIDDLE FINGER BETWEEN THE SKID AND THE ROLLER. LEFT MIDDLE INDEX FINGER BROKEN AT TIP OF FINGER

K-means Only: Cluster Documents Weakly Represented by LDA - 1 of 2

These narratives are included in the K-means finger cluster but are not among the equally sized strongest LDA-topic document set. They may be valid cluster members that use different vocabulary, mixed incidents, or records grouped by K-means because of overall document similarity.

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================================================================================
Example 1 | Record 220012110120 | Days away
K-means cluster: 6 | K-means silhouette: 0.052
Matched LDA topic proportion: 0.124 | Dominant LDA topic: 8
Existing classification: HANDLING OF MATERIALS
Existing accident type: Struck by falling object

APROX 5" STONE WOULD NOT CLIMB BELT, INSTEAD THE STONE KEPT ROLLING AGAINST HOPPER BACKSTOP, EE GRABBED
THE STONE THEN DROPPED IT, THE STONE BOUNCED ONTO HIS FINGER DAMAGING HIS OUTER SKIN.

================================================================================
Example 2 | Record 220071500016 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.056
Matched LDA topic proportion: 0.141 | Dominant LDA topic: 8
Existing classification: HANDTOOLS (NONPOWERED)
Existing accident type: Struck by... NEC

The worker was using a hammer and wedge to split a stone.  The head of the hammer came off the handle
and struck the worker's finger.  The finger split open and and suffered a tuft fracture.

================================================================================
Example 3 | Record 220133440017 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.093
Matched LDA topic proportion: 0.153 | Dominant LDA topic: 8
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

They were rolling a belt onto the Cobra Saw and he got his finger cut and broken.

================================================================================
Example 4 | Record 220043410049 | Days away
K-means cluster: 6 | K-means silhouette: 0.051
Matched LDA topic proportion: 0.156 | Dominant LDA topic: 2
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

EE WAS PUTTING SLIP FLANGES ON 12" BASALT PIPE.  THE FLANGE FELL OFF PIPE AND SMASHED EE'S FINGER.
MEDICAL TREATMENT ONLY.

================================================================================
Example 5 | Record 220022170143 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.038
Matched LDA topic proportion: 0.174 | Dominant LDA topic: 8
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between NEC

EE WAS IN THE PROCESS OF PUTTING A BED PIN IN PLACE ON A HAUL TRUCK. THE PIN SLIPPED AND PINCHED HIS
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K-means Only: Cluster Documents Weakly Represented by LDA - 2 of 2

FINGER BETWEEN THE PIN AND THE LADDER. HE RECEIVED 7 SUTURES AND 4 SHIFTS OF LIGHT DUTY.

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Example 6 | Record 220002940030 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.056
Matched LDA topic proportion: 0.217 | Dominant LDA topic: 8
Existing classification: HANDTOOLS (NONPOWERED)
Existing accident type: Caught in, under or between a moving and a stationary object

EE STATES HE WAS MOVING TREE ROOT WITH A PRY BAR AND SMASHED HIS FINGER BETWEEN BAR AND STUMP. FRACTURE WAS DIAGNOSED AT A LATER DATE.

LDA Only: Potential Documents Missing from the K-means Cluster - 1 of 2

These narratives are outside the selected K-means cluster but have high proportions of the matched LDA topic. These are the primary candidates for narratives that the first-pass K-means cluster may have missed.

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================================================================================
Example 1 | Record 220080360005 | Permanent disability
K-means cluster: 7 | K-means silhouette: -0.024
Matched LDA topic proportion: 0.979 | Dominant LDA topic: 3
Existing classification: POWERED HAULAGE
Existing accident type: Caught in, under or between running or meshing objects

The employee was cleaning out around the Zero Speed hose, while the belt was running. He opened up the guard and stuck his Right hand in, past bearing for the tail pulley, with a pair of Channel locks to clean. When he lost his footing, slipped and fell into the belt, the tail pulley grabbed his channel locks and pulled him into the tail pulley.

================================================================================
Example 2 | Record 220041280050 | Permanent disability
K-means cluster: 8 | K-means silhouette: -0.021
Matched LDA topic proportion: 0.975 | Dominant LDA topic: 3
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between NEC

EE was investigating a noise coming from MC-1 unit, cleaner conveyer tail pulley. He removed the guard from the cleaner conveyer unit and inserted his right hand to check a weld on the pulley to determine if the weld was defective. The tip of the finger was caught and pinched between the bushing and the tail pulley shaft, resulting in loss of fingertip above cuticle.

================================================================================
Example 3 | Record 220110540001 | Permanent disability
K-means cluster: 8 | K-means silhouette: -0.006
Matched LDA topic proportion: 0.975 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

Member was in the deck operating the bolter. He went to reposition himself and stuck his hand on top of the canopy to pull himself up and hit the tram. The bolter moved, smashing his right hand into the top. Member immediately ceased to work and was taken by ambulance to CAMC General where he was told his right hand/middle finger had to be amputated.

================================================================================
Example 4 | Record 220253170158 | Restricted or transferred
K-means cluster: 8 | K-means silhouette: 0.046
Matched LDA topic proportion: 0.972 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

Injured MTM was sitting in a spare 3rd stage mainframe in the zinc shop installing a horse collar liner. EE was operating the overhead crane with EE's right hand and repositioning a horse collar liner for final placement with EE's left hand when EE's left fingertips became pinched between the bottom of the tub and the liner. This resulted in a fractured left middle fingertip
```

LDA Only: Potential Documents Missing from the K-means Cluster - 2 of 2

```
================================================================================
Example 5 | Record 220141350001 | Permanent disability
K-means cluster: 8 | K-means silhouette: -0.000
Matched LDA topic proportion: 0.971 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Caught in, under or between running or meshing objects

The employee was cleaning the rotary vane valve unit on a bag house discharge. The employee placed his
hand into the vane valve housing, at which time the unit activated, rotated, and caught the end of a
finger on his left hand, causing a crushing avulsion.

================================================================================
Example 6 | Record 220041950007 | Days away
K-means cluster: 8 | K-means silhouette: -0.027
Matched LDA topic proportion: 0.970 | Dominant LDA topic: 3
Existing classification: HANDLING OF MATERIALS
Existing accident type: Struck by... NEC

The tailgate on the back of the Mack dump truck would not latch properly.  In order to get tailgate to
latch employee pulled the tailgate up with one hand and lifted the latch with the other.  The tailgate
slipped out of his hand and smashed the middle finger on his left hand.  Employee required 6 stitches.

================================================================================
Example 7 | Record 220062510033 | Permanent disability
K-means cluster: 1 | K-means silhouette: 0.039
Matched LDA topic proportion: 0.970 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

Employee was tramming a roof bolting machine towards the face and turning into a right  crosscut.  He
had his right hand outside of the deck area and when the machine turned his hand was caught between
frame and the coal rib, mashing his right hand.

================================================================================
Example 8 | Record 220162530013 | Days away
K-means cluster: 8 | K-means silhouette: 0.007
Matched LDA topic proportion: 0.970 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Struck by... NEC

Working on C side & had left hand behind bracket & took impact tool to run bolt in. Bolt went through
bracket and smashed left small finger. Cause of injury hand placement. Incident didn't involve any
aspect of compliance with regulations/rules/PPE. Miner proficiency may have been a factor.
```

Borderline Documents within the K-means Finger Cluster - 1 of 2

These selected-cluster narratives have the lowest K-means silhouette values. They may contain several event patterns or may fit another cluster as well as, or better than, the assigned finger cluster.

==========
Example 1 | Record 220013510004 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.021
Matched LDA topic proportion: 0.591 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

EE WAS BOLTING ON MINER. HE WAS ATTEMPTING TO INSERT DRILL STEEL INTO CHUCK AS HE RAISED THE DRILL HEAD. THE STEEL CAUGHT AGAINST THE ROOF & BENT, CATCHING HIS L. LITTLE FINGER BETWEEN THE DRILL STEEL AND THE BOLTER MAST.

==========
Example 2 | Record 220051400016 | Days away
K-means cluster: 6 | K-means silhouette: 0.024
Matched LDA topic proportion: 0.944 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

Member, was moving rt. roof bolter when atrs extension came out its slots. When moving it back in place the end came up smashing finger on roof.

==========
Example 3 | Record 220001540056 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.028
Matched LDA topic proportion: 0.964 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

INJURED WAS ON RIGHT SIDE OF BOLTER WHILE OPERATOR WAS TRAMMING FORWARD FOR NEXT ROW OF BOLTS & BOLTER SLID OVER & GOT ON HIS RIGHT HAND MASHINGTHE ENDS OF HIS RING, MIDDLE & POINTER FINGERS. INJURED WAS TAKEN TO HOSPITAL. THE END OF HIS RING & POINTER FINGER (FROM LAST JOINT) WAS REMOVED.

==========
Example 4 | Record 220031500007 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.030
Matched LDA topic proportion: 0.256 | Dominant LDA topic: 8
Existing classification: HANDTOOLS (NONPOWERED)
Existing accident type: Caught in, under or between NEC

THE MAN WAS LOWERING A RATCHET TYPE LIFTING JACK. THE MAN'S FINGER WAS CAUGHT WHEN THE JACK HEAD DROPPED DOWN. LIGHT WORK DUTY STARTED ON 5/19.

==========
Example 5 | Record 220052020014 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.032
Matched LDA topic proportion: 0.754 | Dominant LDA topic: 3
Existing classification: MACHINERY
Existing accident type: Struck by... NEC

Borderline Documents within the K-means Finger Cluster - 2 of 2

Tightening a spring bolt and was using an air wrench, slipped and cut some of his pinky finger off.

Conclusions and Recommended Interpretation

1. K-means and LDA use the same narratives and vocabulary, but they represent document structure differently.

2. K-means assigns each narrative to exactly one cluster based on overall document similarity in the reduced TF-IDF representation.

3. LDA assigns each narrative a mixture of topic proportions. A narrative can therefore contain the finger topic along with machinery, material-handling, body-part, or injury topics.

4. The methods agreed on 127 narratives. These records provide the strongest evidence for a coherent finger pinch/crush narrative group.

5. K-means contained 120 records not included in the strongest LDA set. These records should be checked for mixed vocabulary, cluster-boundary effects, or weaker finger content.

6. LDA identified 120 records outside the K-means cluster. These records are candidates for documents missing from the first-pass K-means cluster.

7. LDA-only records should not automatically be added to the cluster. They may share finger or hand terms while describing a different mechanism, activity, or injury pattern.

8. The most important qualitative review is whether the LDA-only narratives describe the same abstract concept: finger pinch, crush, caught-between, degloving, or amputation events.

9. If relevant records are consistently separated because of equipment or activity vocabulary, future preprocessing could normalize the broader event mechanism rather than relying only on exact terms.

10. This analysis demonstrates why first-pass clusters need review. Term-level similarity helps group documents, but more abstract safety concepts are needed to connect differently worded events.

Technical Appendix

```
Narratives analyzed: 2,000
Shared vocabulary terms: 2,763
Minimum document frequency: 3
Maximum document frequency: 70%
N-gram range: (1, 2)
Maximum vocabulary size: 8,000
SVD components: 75
SVD explained variance: 0.2270
K-means cluster count: 9
K-means silhouette score: 0.0533
Selected finger cluster: 6
Selected cluster size: 247
Selected cluster mean silhouette: 0.0918
LDA topic count: 9
LDA iterations: 30
Matched LDA topic: 3
Agreement count: 127
K-means-only count: 120
LDA-only count: 120
Document-set Jaccard: 0.3460
Random state: 42

PDF output: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Finger_Cluster_KMeans_LDA_Comparison.pdf
CSV output: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final
Project/MSHA_Finger_Cluster_KMeans_LDA_Document_Comparison.csv
```